Ocean Guard AI Design Project Description Summary

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1 Project Overview

Ocean Guard AI is a large-scale, AI-powered system designed to monitor and protect marine life in real-time. It utilizes underwater drones, sonar arrays, and advanced machine learning models to detect and classify threats to marine life across the ocean.

2 The Purpose of the Project

This project develops an AI-based system which enables marine specialists to monitor endangered species in real-time while generating useful information for worldwide conservation and scientific research.

3 The Scope of the Work

The project focuses on real-time monitoring and alert systems, deterrence methods, and data collection functions, while excluding all other aspects of marine conservation.

4 The Scope of the Product

Ocean Guard AI provides detection and alert systems, safe deterrence, and analytics to support conservation teams, researchers, and fisheries with real-time data and insights for marine life protection.

5 Stakeholders

- The Client: Organizations that provide funding and strategic direction for marine conservation efforts, including WWF, NOAA, and UNEP, which are both government agencies and marine conservation organizations.
- The Customers: The system serves organizations that include conservation groups, fisheries, government agencies, and research institutions for their real-time monitoring and reporting needs.
- Hands-On Users: The system serves scientists who work as marine biologists, conservation teams, fisheries managers, and researchers who need it for detection, analytics, and their regular field operations.
- Maintenance Users: System administrators, developers, and field technicians who perform installation tasks, updates, and maintenance services.
- Other Stakeholders: Sponsors, legal advisors, and international organizations work together to guarantee compliance and worldwide cooperation.
- User Participation: Users participate in testing, AI accuracy evaluation, and design reporting during the entire development process.
- User Priorities: The protection of marine species stands as the main priority for both marine biologists and conservation teams, while research institutions and fisheries follow as the second priority, and the public receives benefits through indirect means.

6 Mandated Constraints

However, building this Ocean Guard AI system, we will face certain obstacles that can halt or slow down the progression of the development. We have to account for multiple different constraints including technology functionality, environmental challenges, device connections, budgeting of the project, and the limited time of development. We have taken the constraints that will come with developing the project into account and thought of some ideas to resolve these conflicts.

7 Naming Conventions and Definitions

This section outlines the fundamental data components together with their established naming conventions which form the basis of Ocean Guard AI operations. The system includes essential entities which consist of Species, Encounters, Alerts, Devices, and Users.

8 Relevant Facts and Assumptions

Ocean Guard AI emerges from the worldwide necessity to develop protective technologies for marine environments. The world lacks any operational system which provides real-time protection for ocean species across all global waters while biodiversity continues to decline because of climate change and human activities. Scientists together with conservationists and government officials work to find technological solutions that will help solve these environmental problems.

The system design depends on the successful operation of underwater drones and sonar arrays in harsh ocean environments and the ability of AI models to identify marine species and detect potential threats. The project must follow international environmental laws and ethical standards to achieve safe and responsible global marine conservation through its implementation.